

S^2KT : Modeling Uncertainty in Knowledge Tracing via Semantic-aware Structured Gaussian Distributions

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Abstract

Knowledge Tracing (KT) aims to track the evolving mastery of Knowledge Components (KCs) and serves as the cornerstone of personalized intelligent tutoring systems. Despite the advancements in deep learning, existing models predominantly rely on static and discrete relational structures. This paradigm is highly susceptible to data biases and fails to capture the continuous, uncertainty-aware semantic dependencies inherent in educational content. Such limitations hinder the precise quantification of latent Knowledge Transfer and oversimplify the complex, multi-scale forgetting dynamics associated with different KCs. Consequently, this exacerbates assessment bias against students with sparse interaction histories or atypical learning patterns. To address these challenges, we propose S^2KT (Semantic-aware Structured Knowledge Tracing). Specifically, we construct a robust semantic manifold using a centrality-guided adaptive graph-walking strategy and represent each KC as a Gaussian Embedding, explicitly capturing both semantic hierarchies and intrinsic uncertainty. To model fine-grained knowledge transfer, we introduce a semantic propagation mechanism based on the Bhattacharyya distance to diffuse mastery updates within the latent space. Furthermore, we propose a Dynamic GMM Attention Module that integrates Item Response Theory (IRT) with Gaussian mixture constraints. This module dynamically synthesizes personalized

learning gains while explicitly modeling heterogeneous forgetting trajectories coupled with KC complexity. Extensive experiments on six real-world datasets demonstrate that S^2KT significantly outperforms 22 state-of-the-art baselines in both accuracy and interpretability.

CCS Concepts

• Applied computing → E-learning.

Keywords

Knowledge Tracing; Semantic-aware Structured; Gaussian Distribution

ACM Reference Format:

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1 Introduction

Knowledge Tracing (KT) is a pivotal paradigm that leverages students' historical interaction data to predict their future performance [13]. Fundamentally cast as a sequence prediction task, KT aims to dynamically model the evolution of students' knowledge mastery. This capability underpins widespread applications in online education, including Massive Open Online Courses (MOOCs) and Intelligent Tutoring Systems (ITS) [25]. By analyzing learning trajectories, KT facilitates the adaptive adjustment of instructional content and pacing, thereby improving learning outcomes [44].

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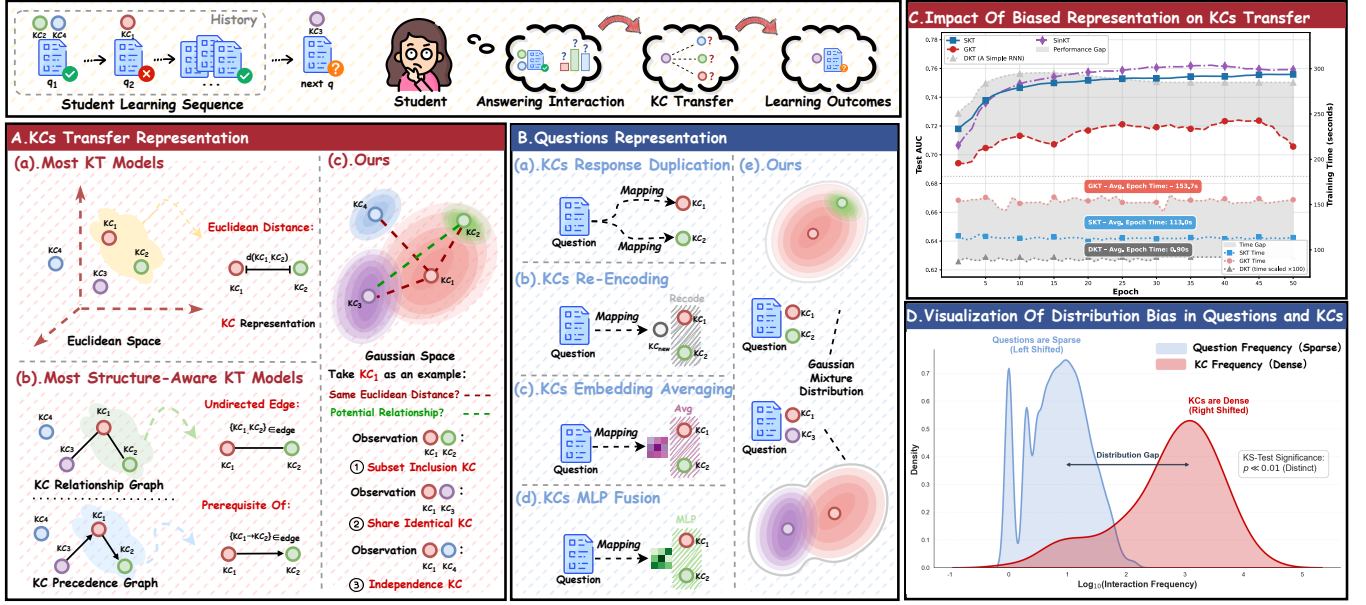


Figure 1: Illustration of the Knowledge Semantic Structure Contained in Questions.

Such adaptability is crucial for advancing next-generation education systems toward intelligence and personalization.

The central objective of KT is to assess a student's proficiency in specific Knowledge Components¹ (KCs) based on their exercise performance [2]. A question is rarely an isolated entity; rather, it represents a specific constellation of KCs. Solving a question requires the **synergistic application** of these KCs, which collectively constitute the cognitive structure underlying the learning task.

In the context of Deep Learning-based KT (DLKT), the pioneering DKT model [42] employed Recurrent Neural Networks (RNNs) [31] to model knowledge states using interaction histories derived from question or KC IDs². Subsequent research has augmented this framework with architectures such as Graph Neural Networks (GNNs) [26] and Transformers [48]. However, these models predominantly rely on discrete ID embeddings, which struggle to capture the complex, inherent semantic structures among KCs. Furthermore, to mitigate the sparsity of student-problem interactions, some approaches simplify the task by assuming a one-to-one mapping between questions and KCs [29, 32]. This oversimplification obscures the rich semantic structure embodied in real-world questions, thereby compromising the fidelity of student knowledge modeling.

Recent studies suggest that an effective KT method must rigorously align with the semantic structures of knowledge [1, 9]. Specifically, this necessitates satisfying two key criteria: (1) **Inter-KC Modeling**: modeling the transfer of learning gains across related domains by explicitly representing the semantic topologies among KCs; and (2) **Intra-Question Modeling**: performing fine-grained modeling of the inherent semantic compositions within questions to capture the comprehensive influence of multiple KCs.

To this end, existing research has attempted to model these aspects for more accurate characterization. Most studies utilize graph

structures to model KC relationships. For example, GKT [38] represents connections via a knowledge graph, while SKT [47] and SinKT [16] employ prerequisite graphs derived from statistical inference or LLMs [3]. However, such methods typically only **implicitly indicate** sequential dependencies (see Fig. 1A(b)), failing to precisely distinguish logical relationships. Crucially, they neglect implicit semantic topologies—such as subordination (**subset inclusion**), attribute-based similarity (**shared attributes**), or **independence** (see Fig. 1A(c))—consequently impeding the quantification of KC associations. On the other hand, to alleviate sparsity and KC diversity challenges, some studies employ embedding fusion strategies, such as aggregating KC embeddings via Multi-Layer Perceptrons (MLPs) or averaging (see Fig. 1B). However, such **naive aggregation** often fails to preserve the distinct content of individual KCs and ignores their structural dependencies, ultimately limiting the representation capability of the student's knowledge state.

To further enhance the quality of student knowledge state modeling, two primary challenges:

Challenge I: How to capture the intrinsic uncertainty and continuous semantic topology among KCs?

Most prevailing KT paradigms rely on measuring KC similarities via discrete, deterministic embeddings. To capture structural dependencies, recent approaches have incorporated binary graph priors. However, our experiments (see Fig. 1C) indicate that these methods are brittle: reliance on low-quality or heuristically defined graphs often introduces significant structural noise, resulting in performance inferior to simple RNN-based models. We attribute this failure to two intrinsic mismatches between existing models and cognitive reality: (1) **Granularity Mismatch**: Concepts vary significantly in semantic scope and ambiguity. Conventional deterministic point embeddings cannot encode these uncertain semantic boundaries, thereby enforcing a rigid, uniform representation on heterogeneous

¹A KC generalizes terms such as concept, principle, fact, or skill.

²An ID serves as a discrete symbolic identifier.

concepts. (2) **Discrete Connectivity**: Learning transfer is a continuous, probabilistic phenomenon, not a binary switch. Static binary graphs fail to capture the nuanced, continuous influence strength between concepts, especially within sparse interaction data. Consequently, a pivotal challenge remains: how to construct a representation that intrinsically supports both uncertainty-aware granularity and continuous semantic propagation.

Challenge II: How to effectively model the joint knowledge acquisition of heterogeneous semantics within questions?

While questions serve as the carriers of knowledge, they exhibit intrinsic heterogeneity relative to their constituent KCs. As illustrated in Fig. 1D, questions and KCs follow two distinct distributions, residing in divergent semantic spaces. This distributional discrepancy implies that the relationship between a question and its KCs cannot be bridged by simple embedding fusion or averaging. Such naive aggregation strategies overlook the complex nature of student knowledge evolution, which is a dynamic process governed by both cumulative learning and memory decay. Consequently, the pivotal challenge lies in designing a mechanism that can adaptively model the dynamic knowledge acquisition for distinct KCs, explicitly accounting for their semantic distributional differences and the resulting heterogeneous forgetting trajectories.

To address these challenges, we propose the Semantic-aware Structured Knowledge Tracing (S²KT) framework, which introduces a Gaussian Mixture Distribution paradigm to model the uncertain and heterogeneous nature of student learning. To address **Challenge I**, we replace rigid point embeddings with Gaussian distributions to encapsulate semantic granularity. The Knowledge Semantic Structure Module constructs a hybrid graph and employs centrality-guided adaptive walks with LSH-based contrastive learning to capture robust topologies. Furthermore, the Semantic Propagation Module models knowledge transfer as a continuous diffusion process over the Gaussian manifold, quantified via the Bhattacharyya distance. To address **Challenge II**, we introduce the Dynamic GMM Attention Module to bridge the distributional gap. This module models the joint influence of multiple KCs as a Gaussian Mixture process, incorporating a multi-scale forgetting mechanism driven by KC-specific variances to capture heterogeneous decay rates and personalized knowledge gains. Extensive experiments on six real-world datasets validate that S²KT effectively reconstructs the semantic structure of KCs and integrates it into knowledge tracing, yielding significant improvements in student performance prediction accuracy.

Overall, we summarize our contributions as follows:

- We propose S²KT, which departs from deterministic embeddings by modeling KCs as Gaussian distributions. This novel approach effectively captures the uncertain semantic boundaries and varying granularity of KCs.
- We utilize adaptive walks and Bhattacharyya distance-based diffusion on a hybrid graph to enable robust and continuous knowledge transfer.
- We design a Dynamic GMM Attention Module with variance-based multi-scale forgetting to model heterogeneous decay rates and joint knowledge acquisition.
- Extensive experiments on six real-world datasets demonstrate that S²KT consistently outperforms 22 competitive baselines.

2 Preliminaries

In this section, we formally define the KT problem. Subsequently, we present the construction of the hybrid KC-Question graph, which serves as the structural foundation for modeling latent semantic dependencies in our proposed framework.

2.1 Problem Formulation

The fundamental objective of KT is to model the dynamic evolution of a student’s knowledge mastery throughout their learning process. Formally, let $\mathcal{S}$, $\mathcal{Q}$, and $\mathcal{C}$ denote the sets of distinct students, exercises, and KCs, respectively. For each student $s \in \mathcal{S}$, the learning trajectory is represented as a temporally ordered sequence of T interactions, denoted as $X_s = \{x_1, x_2, \dots, x_T\}$.

Each interaction is defined as a quadruple $x_t = (q_t, C_t, a_t, \tau_t)$, where $q_t \in \mathcal{Q}$ represents the question attempting at step t ; $C_t \subseteq \mathcal{C}$ denotes the set of KCs associated with q_t (noting that an question may involve multiple KCs); $a_t \in \{0, 1\}$ is a binary indicator where 1 signifies a correct response and 0 otherwise; and τ_t represents the timestamp of the interaction.

Given the historical interaction sequence X_s up to time step T , the goal of the KT task is to estimate the probability that student s will correctly answer a target question q_{T+1} at the next time step. Mathematically, this is formulated as predicting $P(a_{T+1} = 1 \mid X_s, q_{T+1})$.

2.2 Hybrid Graph Construction

To explicitly model the complex dependencies between questions and KCs, we construct a hybrid graph $\mathcal{G}_H = (\mathcal{V}_H, \mathcal{E}_H)$, where the node set $\mathcal{V}_H = \mathcal{Q} \cup \mathcal{C}$ is the union of disjoint question and KC sets. The edge set $\mathcal{E}_H = \mathcal{E}_{sim} \cup \mathcal{E}_{rel}$ synthesizes two distinct types of interactions: (1) Implicit question correlations $\mathcal{E}_{sim} = \{(q_i, q_j) \mid q_i, q_j \in \mathcal{Q}, \text{sim}(q_i, q_j) \geq \theta\}$, where edges are established if the pairwise similarity exceeds a threshold θ ; since ground-truth labels for question relationships are unavailable, we employ a statistical inference method to compute $\text{sim}(\cdot)$ (detailed in Appendix C.1). (2) Explicit question-KC associations $\mathcal{E}_{rel} = \{(q_m, c_n) \mid q_m \in \mathcal{Q}, c_n \in \mathcal{K}_{q_m}\}$, where $\mathcal{K}_{q_m} \subseteq \mathcal{C}$ denotes the specific set of KCs required to solve question q_m .

3 Methodology

In this section, we present the technical framework of S²KT. As illustrated in Figure 2, the model captures the continuous and uncertainty-aware dynamics of student learning through four integral components: 1) *Knowledge Semantic Structure Module* Constructs Gaussian embeddings for KCs via a centrality-guided adaptive graph walk, capturing both semantic hierarchies and inherent uncertainties. 2) *Interactive Embedding Module* Encodes historical student interactions by incorporating IRT-based difficulty parameters and response variability to create granularity-aware representations. 3) *Semantic Propagation Module* Models latent knowledge transfer by diffusing mastery updates across semantically related KCs, quantified by the Bhattacharyya distance in the Gaussian space. 3) *Dynamic GMM Attention Module* Synthesizes personalized knowledge states by aggregating KC features and capturing multi-scale forgetting dynamics via a Gaussian-constrained attention mechanism.

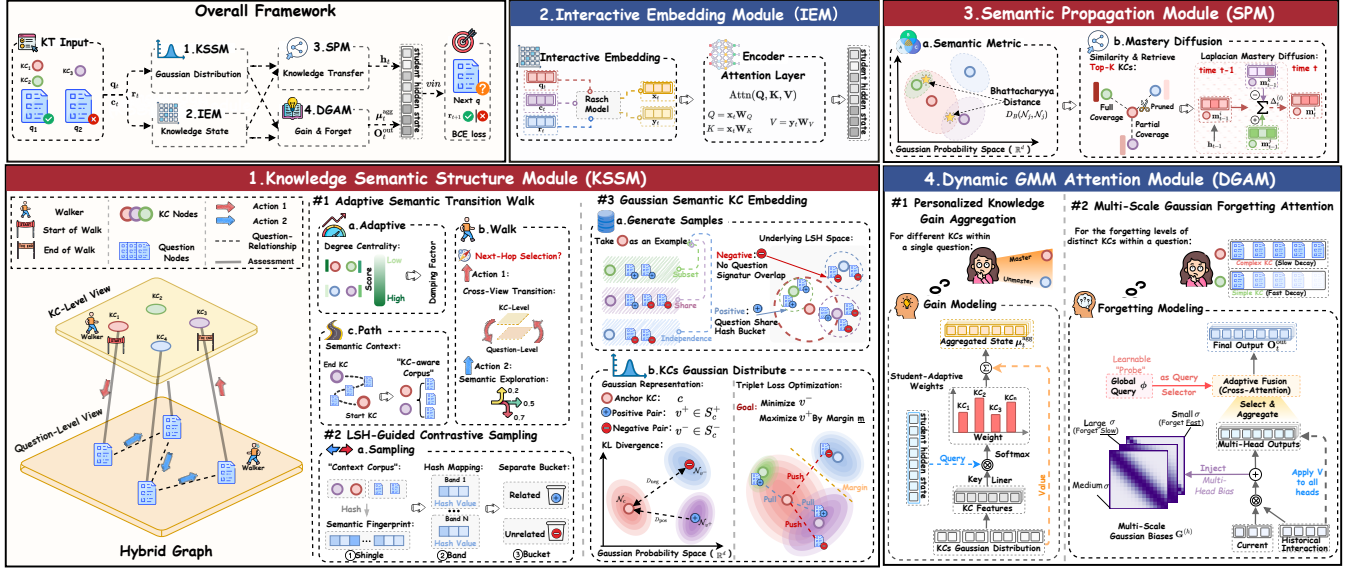


Figure 2: The overall architecture of the proposed S²KT.

3.1 Knowledge Semantic Structure Module

3.1.1 Motivation. Effective KT requires a nuanced understanding of the semantic dependencies among KCs to accurately characterize learning transfer. However, prevailing methods typically derive KC relationships directly from statistical interaction data using simple connectivity measures. This approach suffers from two critical limitations: (1) **Sparsity and Noise:** Interaction data is often sparse and biased, leading to unreliable dependency inference; (2) **Lack of Uncertainty Modeling:** Existing deterministic embeddings fail to capture the inherent ambiguity and varying granularity of different KCs.

3.1.2 Formulation. We propose the Knowledge Semantic Structure Module, which conducts adaptive walks on a hybrid KC-Question graph to learn the semantic context of target KCs, and then embeds KCs as Gaussian distributions to jointly capture semantic similarity and uncertainty.

Adaptive Semantic Transition Walk. Standard random walks (e.g., DeepWalk [41]) often fail to capture the structural nuances of long-tail KCs due to fixed path lengths. We propose a topology-aware strategy that adapts exploration depth based on node importance. Let $\mathcal{G}_c = (V_c, E_c)$ denote the KC subgraph. We compute the normalized degree centrality for each KC node v_i :

$$W_{\text{deg}}(v_i) = \frac{d(v_i)}{|V_c| - 1}, \quad (1)$$

where $d(v_i)$ is the degree. We then define an adaptive **damping factor** λ_i to modulate the walk length:

$$\lambda_i = \lambda_{\min} + (\lambda_{\max} - \lambda_{\min})W_{\text{deg}}(v_i), \quad (2)$$

where $\lambda_{\min}, \lambda_{\max} \in [0, 1]$. Nodes with higher centrality are assigned larger λ_i , enabling extended random walks to capture broader semantic associations.

To model the interplay between KCs and Questions, we utilize a hybrid graph $\mathcal{G}_H$. We define an indicator function $\mathbb{I}_c(v^{(t)})$ which equals 1 if node $v^{(t)}$ is a KC, and 0 otherwise. A semantic path

$\mathcal{P} = (v^{(0)}, \dots, v^{(L)})$ is generated starting from $v^{(0)} \in V_c$, where the path length is dynamically determined by the return to the KC view. The walk involves two stochastic actions:

• **Action 1: Cross-View Transition.** We dynamically control the switching probability between the KC view $\mathcal{G}_c$ and the Question view $\mathcal{G}_q$ to prevent semantic drift. At step t , the transition probabilities are governed by the decay factor λ :

$$P(\text{view}^{(t+1)} = \mathcal{G}_q \mid v^{(t)}) = (\lambda^t)^{\mathbb{I}_c(v^{(t)})}, \quad (3)$$

$$P(\text{view}^{(t+1)} = \mathcal{G}_c \mid v^{(t)}) = 1 - P(\text{view}^{(t+1)} = \mathcal{G}_q \mid v^{(t)}). \quad (4)$$

This exponential decay ensures that walks originating from a KC efficiently explore associated questions before converging back to the KC space, aligning the exploration scope with the structural complexity of the concept.

• **Action 2: Semantic Exploration.** Given the selected view $\mathcal{V} \in \{\mathcal{G}_c, \mathcal{G}_q\}$, the next node u is sampled from the neighbors $\mathcal{N}_{\mathcal{V}}(v^{(t)})$ proportional to edge weights:

$$P(v^{(t+1)} = u \mid v^{(t)}, \mathcal{V}) = \frac{w(v^{(t)}, u)}{\sum_{u' \in \mathcal{N}_{\mathcal{V}}(v^{(t)})} w(v^{(t)}, u')}. \quad (5)$$

LSH-Guided Contrastive Sampling. To efficiently distill robust semantic signals from the generated paths $\mathcal{P}$, we introduce a sampling mechanism based on Locality-Sensitive Hashing (LSH) [18]. This approach filters out noise and disentangles structurally unrelated KCs. For each question node visited, we construct a *shingle* combining the node identity and its walk context. These shingles are mapped to signature vectors and partitioned into bands. Questions colliding in the same LSH bucket are identified as **structural candidates**, implying latent semantic similarity.

For a target KC c_i , we define:

- **Positive Set $S_{c_i}^+$:** Questions co-occurring with c_i in LSH buckets, capturing high-confidence semantic overlaps.
- **Negative Set $S_{c_i}^-$:** Questions with zero signature overlap across all bands, representing the maximal separation in the hash space.

This LSH-filtered sampling ensures that the model learns from "hard" positives and distinct negatives, significantly enhancing discriminative power.

Gaussian Semantic KC Embedding. We embed each KC c_i as a multivariate Gaussian distribution $\mathcal{N}_{c_i}(\mu_i, \Sigma_i)$ to simultaneously encode its semantic centroid and uncertainty (breadth) [49]. The asymmetry of knowledge transfer is naturally modeled by the Kullback-Leibler (KL) divergence [27]:

$$\mathcal{D}_{KL}(\mathcal{N}_i \parallel \mathcal{N}_j) = \frac{1}{2} \left(\text{tr}(\Sigma_j^{-1} \Sigma_i) + (\mu_i - \mu_j)^\top \Sigma_j^{-1} (\mu_i - \mu_j) - d - \ln \frac{\det(\Sigma_j)}{\det(\Sigma_i)} \right). \quad (6)$$

To optimize these representations, we employ a margin-based triplet loss. For an anchor KC c , a positive sample $v^+ \in S_c^+$, and a negative sample $v^- \in S_c^-$, the objective enforces that semantically related pairs are closer in the distributional space than unrelated ones:

$$\mathcal{L}_{\text{trip}}(c, v^+, v^-) = \max(0, m + \mathcal{D}_{KL}(\mathcal{N}_c \parallel \mathcal{N}_{v^+}) - \mathcal{D}_{KL}(\mathcal{N}_c \parallel \mathcal{N}_{v^-})), \quad (7)$$

$$O_{\text{KSSM}} = \sum_{c \in \mathcal{V}_c} \mathbb{E}_{v^+ \sim S_c^+, v^- \sim S_c^-} [\mathcal{L}_{\text{trip}}(c, v^+, v^-)]. \quad (8)$$

This objective pulls structurally similar concepts together while pushing dissimilar ones apart, shaping a meaningful probabilistic semantic manifold.

3.2 Interactive Embedding Module

To encode a student's historical learning data, we embed each past question–response pair with fine granularity. Inspired by item–response theory (IRT) and the Rasch model [43], we include a learnable difficulty parameter for each question. Concretely, at time t we define the question embedding $\mathbf{x}_t$ and the question–answer embedding $\mathbf{y}_t$ as:

$$\mathbf{x}_t = \mathbf{e}_{c_t} \oplus (\mathbf{d}_{q_t} \odot \mathbf{v}_{c_t}), \quad (9)$$

$$\mathbf{y}_t = \mathbf{c}_{a_t} \oplus (\mathbf{d}_{q_t} \odot \mathbf{v}_{c_{a_t}}), \quad (10)$$

where $\mathbf{e}_{c_t} \in \mathbb{R}^d$ is the embedding of the current KC c_t , $\mathbf{d}_{q_t} \in \mathbb{R}$ is the learnable difficulty scalar of question q_t , and $\mathbf{v}_{c_t} \in \mathbb{R}^d$ captures item-specific variability for concept c_t (analogous to Rasch item parameters). The operator $\odot$ denotes scalar–vector multiplication and $\oplus$ denotes vector concatenation.

$$\mathbf{c}_{a_t} = \mathbf{e}_{c_t} \oplus \mathbf{e}_{a_t}, \quad \mathbf{v}_{c_{a_t}} = \mathbf{e}_{c_t} \oplus \mathbf{v}_{a_t}, \quad (11)$$

combining the KC embedding $\mathbf{e}_{c_t}$ with the embedding $\mathbf{e}_{a_t} \in \mathbb{R}^d$ of the student's response a_t (e.g. correct/incorrect). where $\mathbf{v}_{a_t} \in \mathbb{R}^d$ captures variability of the student response.

In summary, $\mathbf{x}_t$ embeds the question with its difficulty-adjusted concept features, while $\mathbf{y}_t$ embeds the KC–answer pair. These embeddings form the query/key and value inputs to an attention layer, we compute the student's hidden state:

$$\mathbf{H} = \text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax} \left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} \right) \mathbf{V}. \quad (12)$$

where $\mathbf{Q} = \mathbf{x}_t \mathbf{W}_Q$, $\mathbf{K} = \mathbf{x}_t \mathbf{W}_K$ and $\mathbf{V} = \mathbf{y}_t \mathbf{W}_V$. This attention mechanism aggregates information from all past question–response embeddings, capturing the evolving knowledge state of the student.

3.3 Semantic Propagation Module

3.3.1 Motivation. KT frameworks frequently model KCs as independent entities, ignoring the latent semantic dependencies described in educational psychology (i.e., learning transfer) [5]. While recent graph-based approaches attempt to incorporate KC associations, they typically rely on discrete, binary topological structures. These static representations fail to capture the continuous and uncertainty-aware nature of knowledge transfer, where the strength of influence varies dynamically based on semantic overlap.

3.3.2 Formulation. To bridge this gap, we propose a propagation mechanism that quantifies transfer effects using probabilistic embeddings, enabling dense, weighted interactions between related KCs.

Gaussian Semantic Metric. To capture the semantic correlation between two KCs, c_i and c_j , we leverage their representations in the Gaussian latent space. Let $\mathcal{N}_{c_i} = \mathcal{N}(\mu_i, \Sigma_i)$ and $\mathcal{N}_{c_j} = \mathcal{N}(\mu_j, \Sigma_j)$ denote the Gaussian embeddings for c_i and c_j , respectively. We employ the Bhattacharyya distance [6], a robust metric for measuring the separability of probability distributions, defined as:

$$D_B(\mathcal{N}_i, \mathcal{N}_j) = \frac{1}{8} (\mu_i - \mu_j)^\top \Sigma^{-1} (\mu_i - \mu_j) + \frac{1}{2} \ln \frac{|\Sigma|}{\sqrt{|\Sigma_i| |\Sigma_j|}}, \quad (13)$$

where $\Sigma = \frac{1}{2} (\Sigma_i + \Sigma_j)$. The **first term** is essentially the squared Mahalanobis distance between the means, weighted by the average covariance: it grows when the mean KCs differ in the shared latent space. The **second term** compares the volumes of the covariance ellipsoids: if one KC is much broader or narrower than the other, this term increases. In sum, D_B captures both how far apart the KC centers are and how their uncertainties differ. Small D_B implies the two Gaussian embeddings overlap substantially, indicating semantic similarity. We transform this distance into a bounded similarity score $\text{Sim}_{ij} \in (0, 1]$:

$$\text{Sim}_{ij} = \exp(-D_B(\mathcal{N}_{c_i}, \mathcal{N}_{c_j})). \quad (14)$$

To mitigate noise and improve computational efficiency, we construct a sparse semantic topology by retaining only the top- K nearest neighbors for each KC, denoted as the set $\mathcal{N}_K(i)$.

Laplacian Mastery Diffusion. We model learning transfer as a continuous diffusion process in the logit space. Let m_{t-1}^k denote the mastery logit of KC k prior to the current time step. The updated mastery state m_t^i is derived by aggregating semantic feedback from its neighborhood, analogous to the heat equation on graphs:

$$\Delta_t^{(i)} = \sum_{j \in \mathcal{N}(i, K)} \text{Sim}(c_i, c_j) \cdot (m_{t-1}^j - m_{t-1}^i), \quad (15)$$

$$m_t^i = m_{t-1}^i + \Delta_t^{(i)}. \quad (16)$$

The term $(m_{t-1}^j - m_{t-1}^i)$ acts as a driving force for equilibrium: if a related neighbor c_j has a higher mastery level, it exerts a positive "pull" on c_i , simulating positive transfer. Conversely, significantly lower mastery in related KCs may dampen the estimation, reflecting prerequisite deficiencies. This mechanism allows the model to coherently propagate mastery updates across the semantic manifold.

3.4 Dynamic GMM Attention Module

3.4.1 Motivation. KT models often treat questions and their constituent KCs as independent entities due to distributional differences. This assumption of independence impedes the model’s ability to capture a student’s integrated understanding and the joint application of multiple KCs when solving questions. Additionally, learners exhibit significant individual variability in mastery and forgetting trajectories across different KCs [14]. In particular, different KCs may entail varying depths of understanding and distinct memory decay patterns.

3.4.2 Formulation. To address these challenges, we model the personalized influence of each question on an individual student’s knowledge state through an architecture composed of two main components:

Personalized Knowledge Gain Aggregation. We encode the distributional parameters of each KC as feature vectors $\mathbf{f}_{c_k} = \text{ReLU}([\mu_{c_k}; \sigma_{c_k}^2])$ for $k = 1, \dots, K$. To capture the student’s varying mastery levels across different concepts, we employ an attention mechanism to generate **student-adaptive aggregation weights**. Specifically, we query the KC features using the student’s current hidden state h_t :

$$e_{t,k} = \frac{\mathbf{W}_Q h_t (\mathbf{W}_K \mathbf{f}_{c_k})^\top}{\sqrt{d_h}}, \quad (17)$$

where $\mathbf{W}_Q$ and $\mathbf{W}_K$ are learnable projection matrices. We then normalize these scores to obtain the personalized relevance weights $\pi_{t,k}$ via a temperature-scaled Softmax, and compute the weighted aggregation:

$$\pi_{t,k} = \frac{\exp(e_{t,k}/\tau)}{\sum_{j=1}^K \exp(e_{t,j}/\tau)}, \quad \mu_t^{\text{agg}} = \sum_{k=1}^K \pi_{t,k} \mathbf{f}_{c_k}. \quad (18)$$

This formulation ensures that the aggregated knowledge gain μ_t^{agg} is dynamically modulated by the student’s specific state h_t .

Multi-Scale Gaussian Forgetting Attention. To explicitly model the heterogeneous forgetting dynamics arising from the **varying uncertainties of different KCs** within a question, we construct a dynamic temporal bias matrix $\mathbf{G}_t^{(h)}$. Unlike standard approaches that assume uniform decay, our mechanism drives forgetting rates directly via the specific variance properties of the KCs constituting question q_t :

$$G_{ij,t}^{(h)} = \exp\left(-\frac{(j-i)^2}{2(\sigma_{t,h})^2}\right), \quad (19)$$

where $\sigma_{t,h}$ is the decay bandwidth for the h -th head, derived by projecting the variances $\{\Sigma_c \mid c \in \mathcal{K}_{q_t}\}$ of the question’s constituent KCs into H latent subspaces. Conceptually, this allows the model to disentangle the forgetting dynamics of a single question: KCs with small variances (elementary concepts) trigger rapid decay (narrow Gaussian windows), while KCs with large variances (complex, broad concepts) induce slower decay (wide Gaussian windows).

$$\mathbf{A}^{(h)} = \text{Softmax}\left(\frac{\mathbf{Q}^{(h)} (\mathbf{K}^{(h)})^\top}{\sqrt{d_k}} + \ln(\mathbf{G}^{(h)})\right), \quad (20)$$

$$\mathbf{O}_t^{(h)} = \mathbf{A}_t^{(h)} \mathbf{V}^{(h)}. \quad (21)$$

To adaptively synthesize these diverse forgetting patterns, we deploy a global query mechanism. A learnable vector $\phi \in \mathbb{R}^{d_k}$ probes the semantic validity of each scale’s output $\mathbf{O}_t^{(h)}$, weighing the contribution of each head based on how well its specific variance-driven window captures the relevant historical context. The aggregation weights are computed via a block-aware cross-attention mechanism:

$$\mathbf{u}_{t,h} = \text{MLP}\left(\text{CrossAttn}(\phi, \mathbf{O}_t^{(h)}, \mathbf{O}_t^{(h)})\right), \quad (22)$$

We normalize these weights and derive the final representation:

$$\alpha_{t,h} = \frac{\exp(\mathbf{u}_{t,h}/\tau)}{\sum_{j=1}^H \exp(\mathbf{u}_{t,j}/\tau)}, \quad \mathbf{O}_t^{\text{out}} = \sum_{h=1}^H \alpha_{t,h} \cdot \mathbf{O}_t^{(h)}, \quad (23)$$

where τ is the temperature factor. This formulation enables S^2KT to dynamically attend to the most informative temporal scale, guided specifically by the **intrinsic variances of the KCs** involved in the current question.

3.5 Prediction and Optimization

3.5.1 Formulation. To predict student performance on q_{t+1} , we construct a comprehensive fusion vector $\mathbf{v}_{\text{in}}$ by concatenating multi-view features: the historical state h_t , question embedding q_{t+1} , forgetting context $\mathbf{O}_t^{\text{out}}$, and personalized KC aggregation μ_{t+1}^{agg} :

$$\mathbf{v}_{\text{in}} = h_t \oplus q_{t+1} \oplus \mathbf{O}_t^{\text{out}} \oplus \mu_{t+1}^{\text{agg}}. \quad (24)$$

The probability of a correct response $\hat{r}_{t+1}$ is predicted via a multi-layer perceptron. With the KSSM semantic representations **pre-trained**, we optimize the sequential prediction parameters by minimizing the standard Binary Cross-Entropy (BCE) loss:

$$\hat{r}_{t+1} = \sigma(\mathbf{W}_2 \cdot \phi(\mathbf{W}_1 \mathbf{v}_{\text{in}} + \mathbf{b}_1) + b_2), \quad (25)$$

$$\mathcal{L} = - \sum_t (r_{t+1} \log \hat{r}_{t+1} + (1 - r_{t+1}) \log(1 - \hat{r}_{t+1})), \quad (26)$$

where r_{t+1} denotes the ground truth label.

4 Experiment

In this section, we design comprehensive experiments to address the following research questions(**RQs**):

- **RQ1:** Does S^2KT achieve superior prediction performance while maintaining meaningful KT interpretability? compared to current state-of-the-art DLKT methods?
- **RQ2:** How does each designed component contribute to the effectiveness of S^2KT ?
- **RQ3:** Can S^2KT effectively model interpretable semantic structural relationships between KCs?
- **RQ4:** Does S^2KT achieve a superior trade-off between predictive accuracy and computational efficiency?

Additionally, we conduct other experiments such as empirical analysis of LLM-driven semantic inference. Due to space constraints, we include them in Appendix F.

4.1 Experimental Setup

Datasets & Metrics. We evaluate S^2KT on six widely used real-world benchmarks: ASSIST09, ASSIST12, Algebra05, Junyi, Slepemapy, and Ednet. Detailed statistics and preprocessing steps are provided in Appendix D. We adopt *Accuracy* (ACC) and *Area Under the ROC Curve* (AUC) as evaluation metrics.

Table 1: Overall performance comparison. Bold scores represent the highest results among all the methods, while the second highest scores are underlined. The symbol “+” denotes models that explicitly incorporate knowledge semantic structures, and “-” denotes models that consider the forgetting effect.

Dataset Metric	Assistent09		Assistent12		AL2005		Junyi		Slepemapy		Ednet	
	AUC ↑	ACC ↑	AUC ↑	ACC ↑	AUC ↑	ACC ↑	AUC ↑	ACC ↑	AUC ↑	ACC ↑	AUC ↑	ACC ↑
Sequential KT Models												
DKT	0.7571 ±0.0003	0.7296 ±0.0002	0.7207 ±0.0004	0.7299 ±0.0001	0.8153 ±0.0003	0.8088 ±0.0002	0.7510 ±0.0004	0.8388 ±0.0001	0.7400 ±0.0002	0.8105 ±0.0003	0.6607 ±0.0001	0.6614 ±0.0004
DKT Plus	0.7614 ±0.0002	0.7327 ±0.0003	0.7241 ±0.0001	0.7315 ±0.0002	0.8188 ±0.0005	0.8110 ±0.0002	0.7506 ±0.0003	0.8383 ±0.0002	0.7414 ±0.0004	0.8122 ±0.0001	0.6643 ±0.0003	0.6664 ±0.0002
AT-DKT	0.7544 ±0.0004	0.7259 ±0.0002	0.7173 ±0.0003	0.7278 ±0.0004	0.8136 ±0.0002	0.8077 ±0.0001	0.7500 ±0.0003	0.8382 ±0.0002	0.7401 ±0.0004	0.8115 ±0.0003	0.6613 ±0.0002	0.6642 ±0.0005
ATKT	0.7344 ±0.0003	0.7175 ±0.0004	0.7029 ±0.0001	0.7197 ±0.0003	0.7925 ±0.0005	0.7985 ±0.0002	0.7360 ±0.0002	0.8329 ±0.0001	0.6906 ±0.0005	0.7983 ±0.0003	0.6297 ±0.0001	0.6519 ±0.0004
KQN	0.7546 ±0.0002	0.7288 ±0.0004	0.7206 ±0.0003	0.7300 ±0.0002	0.8099 ±0.0003	0.8063 ±0.0004	0.7513 ±0.0001	0.8394 ±0.0003	0.7419 ±0.0002	0.8103 ±0.0004	0.6669 ±0.0005	0.6652 ±0.0002
QIKT	0.7858 ±0.0004	0.7438 ±0.0001	0.7486 ±0.0003	0.7418 ±0.0002	0.8313 ±0.0005	0.8204 ±0.0002	0.7910 ±0.0004	0.8433 ±0.0001	0.7422 ±0.0002	0.8118 ±0.0003	0.7159 ±0.0002	0.6913 ±0.0004
Memory-Augmented KT Models												
DKVMN	0.7533 ±0.0003	0.7274 ±0.0002	0.7183 ±0.0002	0.7313 ±0.0004	0.7990 ±0.0001	0.8013 ±0.0003	0.7508 ±0.0005	0.8395 ±0.0002	0.7427 ±0.0004	0.8098 ±0.0001	0.6608 ±0.0002	0.6634 ±0.0003
SKVMN +	0.7409 ±0.0002	0.7203 ±0.0004	0.7095 ±0.0001	0.7210 ±0.0003	0.7905 ±0.0002	0.7952 ±0.0002	0.7390 ±0.0004	0.8288 ±0.0001	0.7315 ±0.0005	0.8008 ±0.0003	0.6488 ±0.0002	0.6562 ±0.0001
Deep-IRT	0.7561 ±0.0001	0.7285 ±0.0003	0.7178 ±0.0004	0.7309 ±0.0002	0.7873 ±0.0005	0.8447 ±0.0004	0.7499 ±0.0002	0.8394 ±0.0003	0.7403 ±0.0001	0.8090 ±0.0004	0.6595 ±0.0002	0.6624 ±0.0003
Graph-Based KT Models												
GKT +	0.7371 ±0.0003	0.7218 ±0.0001	0.7182 ±0.0002	0.7296 ±0.0004	0.8012 ±0.0005	0.8021 ±0.0002	0.7385 ±0.0001	0.8290 ±0.0002	0.7255 ±0.0004	0.7985 ±0.0003	0.6455 ±0.0005	0.6501 ±0.0002
SKT +	0.7563 ±0.0004	0.7321 ±0.0002	0.7224 ±0.0001	0.7312 ±0.0003	0.8141 ±0.0002	0.8085 ±0.0004	0.7525 ±0.0005	0.8402 ±0.0003	0.7415 ±0.0001	0.8112 ±0.0002	0.6592 ±0.0004	0.6610 ±0.0001
PEBG	0.7583 ±0.0001	0.7312 ±0.0003	0.7258 ±0.0002	0.7325 ±0.0004	0.8192 ±0.0005	0.8118 ±0.0002	0.7538 ±0.0003	0.8405 ±0.0001	0.7432 ±0.0001	0.8130 ±0.0005	0.6655 ±0.0003	0.6672 ±0.0002
SinKT +	0.7773 ±0.0003	0.7343 ±0.0004	0.7455 ±0.0002	0.7384 ±0.0002	0.8291 ±0.0001	0.8169 ±0.0003	0.7812 ±0.0004	0.8413 ±0.0001	0.7449 ±0.0002	0.8127 ±0.0005	0.7212 ±0.0004	0.6905 ±0.0003
Attention-based KT Models												
SAKT	0.7356 ±0.0002	0.7129 ±0.0003	0.6980 ±0.0004	0.7197 ±0.0002	0.7831 ±0.0005	0.7948 ±0.0002	0.7265 ±0.0001	0.8324 ±0.0004	0.7165 ±0.0003	0.8088 ±0.0005	0.6561 ±0.0002	0.6610 ±0.0003
AKT +	0.7850 ±0.0005	0.7433 ±0.0002	0.7457 ±0.0001	0.7409 ±0.0004	0.8241 ±0.0003	0.8059 ±0.0002	0.7908 ±0.0005	0.8460 ±0.0003	0.7502 ±0.0004	0.8108 ±0.0002	0.7154 ±0.0001	0.6901 ±0.0003
SimpleKT	0.7804 ±0.0002	0.7419 ±0.0004	0.7414 ±0.0005	0.7364 ±0.0001	0.8245 ±0.0002	0.8129 ±0.0003	0.7837 ±0.0004	0.8423 ±0.0005	0.7493 ±0.0001	0.8131 ±0.0003	0.7126 ±0.0002	0.6885 ±0.0004
DTransformer	0.7778 ±0.0004	0.7391 ±0.0002	0.7346 ±0.0003	0.7348 ±0.0005	0.8087 ±0.0001	0.8053 ±0.0004	0.7855 ±0.0002	0.8432 ±0.0003	0.7440 ±0.0005	0.8116 ±0.0001	0.7197 ±0.0004	0.6918 ±0.0002
FluCT +	0.7895 ±0.0003	0.7456 ±0.0005	0.7453 ±0.0002	0.7382 ±0.0004	0.8237 ±0.0003	0.8102 ±0.0001	0.7914 ±0.0005	0.8455 ±0.0003	0.7505 ±0.0002	0.8127 ±0.0004	0.7237 ±0.0001	0.6916 ±0.0005
FoLiBiKT +	0.7899 ±0.0004	0.7443 ±0.0002	0.7454 ±0.0003	0.7408 ±0.0001	0.8230 ±0.0004	0.8102 ±0.0003	0.7923 ±0.0002	0.8461 ±0.0004	0.7510 ±0.0002	0.8126 ±0.0002	0.7233 ±0.0003	0.6958 ±0.0001
SparseKT	0.7772 ±0.0002	0.7367 ±0.0005	0.7377 ±0.0004	0.7354 ±0.0002	0.8231 ±0.0003	0.8116 ±0.0001	0.7757 ±0.0004	0.8395 ±0.0002	0.7388 ±0.0001	0.8098 ±0.0003	0.7150 ±0.0005	0.6891 ±0.0002
StableKT	0.7866 ±0.0001	0.7456 ±0.0004	0.7421 ±0.0003	0.7397 ±0.0002	0.8292 ±0.0005	0.8121 ±0.0003	0.7857 ±0.0002	0.8441 ±0.0004	0.7466 ±0.0001	0.8119 ±0.0002	0.7148 ±0.0003	0.6911 ±0.0005
RobustKT	0.7895 ±0.0003	0.7443 ±0.0002	0.7472 ±0.0005	0.7428 ±0.0001	0.8293 ±0.0004	0.8137 ±0.0003	0.7934 ±0.0002	0.8459 ±0.0001	0.7454 ±0.0003	0.8117 ±0.0004	0.7127 ±0.0001	0.6866 ±0.0003
Proposed Method												
S ² KT	0.7949 ±0.0004	0.7548 ±0.0001	0.7518 ±0.0003	0.7472 ±0.0002	0.8378 ±0.0003	0.8242 ±0.0001	0.8238 ±0.0003	0.8497 ±0.0002	0.7663 ±0.0002	0.8161 ±0.0002	0.7332 ±0.0002	0.7026 ±0.0001
%Improv.	0.63%	1.23%	0.43%	0.73%	0.78%	0.46%	3.83%	0.43%	2.04%	0.42%	1.31%	0.98%

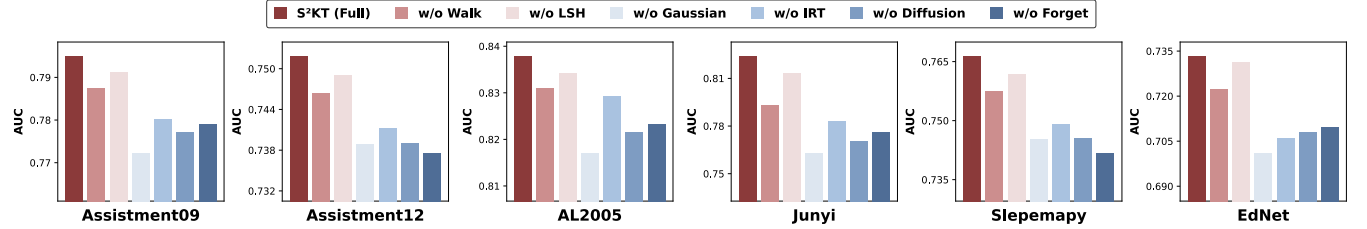


Figure 3: Performance comparison of removing each designed component.

Baselines & Implementation Details. To demonstrate the effectiveness of S²KT, we compare it against 22 competitive methods across four categories: Sequential (e.g., DKT [42]), Graph-based (e.g., GKT [38]), Memory-augmented (e.g., DKVMN [54]), and Attention-based methods (e.g., SimpleKT [34]). A comprehensive list and descriptions of these baselines are detailed in Appendix E.1. Regarding implementation, all models are developed using PyTorch on NVIDIA A800 GPUs with 5-fold cross-validation. For S²KT, we optimize hyperparameters via grid search, with detailed settings provided in Appendix E.2. To facilitate reproducibility, our code and datasets are available at <https://github.com/NEUwwh/S2KT>.

4.2 Overall Comparison (RQ1)

Table 1 reports the performance comparison between S²KT and 22 baselines across six datasets. S²KT consistently yields the best results across all metrics. Our key observations are: (1) **New State-of-the-art Performance:** S²KT outperforms the best baselines with an average AUC improvement of 1.50% (a significant margin considering the field’s saturation, where rigorous benchmarks [35] estimate the cumulative gain since 2015 at only ~3.5% under unified protocols). Notably, on the Junyi dataset, which is characterized by semantic structural dependencies, our model achieved a significant improvement of 3.83%. These results validate that our semantic structural modeling effectively captures fine-grained

knowledge transfer and concept interdependencies. (2) **Superiority over structure-aware Methods:** While structure-aware models (e.g., GKT) attempt to model transfer via explicit relations, they lag behind S²KT by an average AUC gap of ~7.5%. Their performance fluctuation suggests sensitivity to the quality of pre-defined structures. In contrast, the adaptive walk-based modeling in S²KT captures latent dynamics robustly without relying on static structural priors. (3) **Robustness Across Data Scales:** S²KT demonstrates strong generalization, maintaining superiority on both small-scale (Assistent09) and massive, sparse datasets (Ednet, Junyi). Unlike traditional sequential models (e.g., DKT) that degrade on complex distributions, S²KT handles data heterogeneity by integrating semantic structures with sequence modeling. (4) **Semantic-Enhanced Attention:** Although attention-based baselines like SimpleKT are competitive, they often overlook intrinsic semantic logic. S²KT addresses this by injecting semantic structural constraints into the attention mechanism. This effectively balances the capture of long-term sequential patterns with local semantic consistency, fully exploiting the Transformer architecture.

4.3 Ablation Study (RQ2)

We compare S²KT against six variants to validate its components: w/o Adaptive Walk (uses random walks); w/o LSH (uses random sampling); w/o Gaussian Emb. (uses deterministic vectors); w/o IRT (uses one-hot encoding); w/o Diffusion (removes KC dependencies); and w/o Forget (removes the forgetting mechanism). The

performance comparison of removing each of these designed components is illustrated in Figure 3: **(1) Gaussian Structure:** The significant drop in *w/o Gaussian Emb.* confirms that deterministic vectors fail to capture semantic relationships. Similarly, declines in *w/o Adaptive Walk* and *w/o LSH* validate the need for centrality-guided context and effective sampling. **(2) Propagation:** The poor performance of *w/o Diffusion*, especially on the structure-rich Junyi dataset, underscores the necessity of Laplacian diffusion for propagating updates to alleviate sparsity. **(3) Dynamics:** The gap in *w/o Forget*, notably on temporal datasets (Slepemapy), indicates that the variant without explicit decay cannot capture differentiated decay rates, whereas our Gaussian approach models realistic cognitive evolution. **(4) Granularity:** *w/o IRT* lags behind, confirming that capturing item difficulty is essential for precise student profiling beyond generic KC labels.

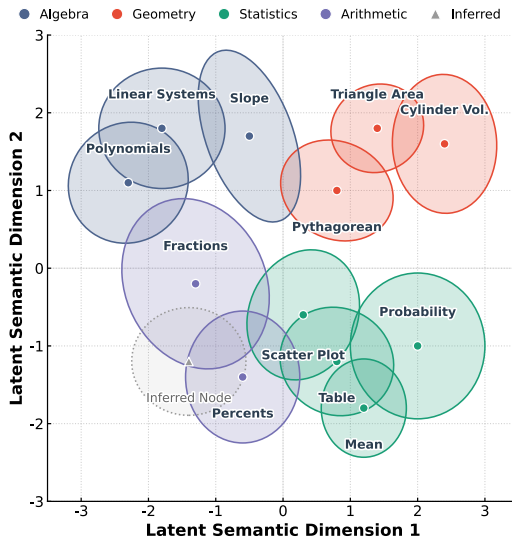


Figure 4: Visualization of the Gaussian Semantic Space.

4.4 Visualization of Semantic Structure (RQ3)

Figure 4 illustrates the 2D projection of 13 sampled KCs, revealing: **(1) Semantic Cohesion:** KCs spontaneously cluster by pedagogical category (e.g., *Geometry*, *Algebra*) without supervision, verifying the module’s ability to preserve semantic topology. **(2) Probabilistic Granularity:** The Gaussian (μ, Σ) representation transcends point-vector limitations. It resolves Euclidean distance ambiguities via distributional overlap, quantifies knowledge sharing through ellipsoid volume, and intrinsically captures hierarchical entailment—where foundational concepts (large variance) spatially encapsulate specific ones. **(3) Cross-Domain Transfer:** “Soft” boundaries reveal latent intersections between disparate topics, acting as bridges for mastery propagation across shared capabilities. **(4) Robustness:** The structure-based encoding resolves polysemy (e.g., context-dependent labels) and enables zero-shot semantic inference for unannotated concepts via graph propagation.

4.5 Efficiency Analysis (RQ4)

To assess the deployability of S^2KT in real-world educational scenarios, we investigate the critical trade-off between predictive effectiveness and computational cost. We benchmark our method against representative baselines that utilize either semantic-driven

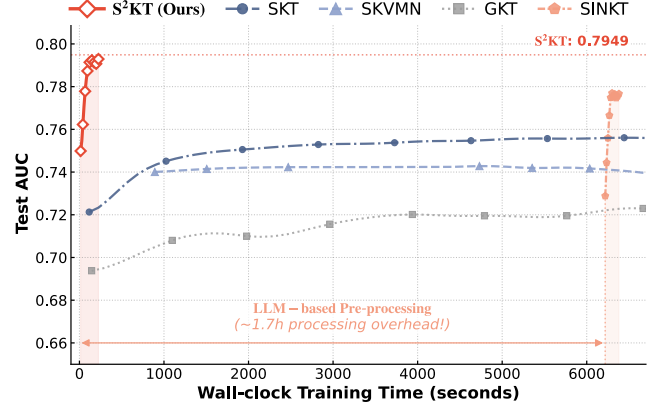


Figure 5: Training Efficiency and Convergence Analysis.

or structure-based modeling paradigms. To ensure a fair comparison, we conduct all efficiency evaluations on the ASSIST09 dataset, a common benchmark shared across the original studies of the selected baselines. We strictly adhere to the hyperparameter settings reported in the respective papers. All models are evaluated on an identical platform (single NVIDIA A800 GPU, batch size 32) to ensure hardware consistency. As illustrated in Figure 5, S^2KT achieves a superior Pareto frontier in the efficiency-accuracy trade-off. The experimental results highlight three key advantages: **(1) Rapid Convergence:** S^2KT achieves significantly faster convergence via the Semantic Propagation Module. By leveraging Laplacian diffusion, mastery updates effectively propagate to related KCs without requiring exhaustive interactions, accelerating learning even in sparse data scenarios. **(2) Superior Effectiveness:** Unlike deterministic point embeddings, S^2KT employs Gaussian Semantic Embeddings to capture both concept centroid and uncertainty. Coupled with Dynamic GMM Attention, this probabilistic modeling enables nuanced distinction of student mastery, yielding higher predictive accuracy. **(3) High Computational Efficiency:** S^2KT maintains high throughput through two key designs: LSH-guided sampling which prunes the semantic search space, and closed-form Gaussian metric computations. These mechanisms avoid the heavy overhead typical of large-scale Graph or LLM-based approaches.

5 Conclusion

In this paper, we proposed S^2KT , a novel Semantic-aware Structured Knowledge Tracing framework that fundamentally shifts the representation paradigm from deterministic point embeddings to probabilistic Gaussian distributions. Through centrality-guided adaptive walks and Bhattacharyya distance-based diffusion, our model effectively captures knowledge uncertainty and continuous semantic dependencies. Moreover, the integration of a multi-scale forgetting mechanism allows for a nuanced characterization of personalized learning trajectories. Empirical results across six datasets confirm that S^2KT outperforms current SOTA methods in accuracy while maintaining high computational efficiency.

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A Notation Table

We list and explain the key notations used in our methodology in Table 2.

Table 2: Summary of Notations

Symbol	Description
<i>Problem Formulation & Graph Construction</i>	
S, Q, C	Sets of Students, Questions, and KCs
X_s	Learning interaction sequence for student s
x_t	Interaction tuple at step t : (q_t, c_t, a_t, τ_t)
$\mathcal{G}_H$	Hybrid graph $(\mathcal{V}_H, \mathcal{E}_H)$
$\mathcal{E}_{sim}, \mathcal{E}_{rel}$	Edge sets: Q-Q similarity & Q-KC association
<i>Knowledge Semantic Modeling</i>	
$W_{deg}(v); \lambda_v$	Norm. degree centrality; Adaptive damping
$\mathbb{I}_c(\cdot)$	Indicator function for KC view distinction
S_c^+, S_c^-	Positive/Negative sample sets via LSH
$\mathcal{N}_c(\mu, \Sigma)$	Gaussian embedding for KC c
$\mathcal{D}_{KL}$	Kullback-Leibler divergence metric
$\mathcal{L}_{trip}$	Margin-based triplet loss function
$D_B(\cdot, \cdot)$	Bhattacharyya distance metric
<i>Dynamic Learning State Modeling</i>	
d_{qt}	Learnable difficulty parameter for q_t
v_{ct}	Item-specific variability vector
x_t, y_t	Embeddings for Question / QA-pair
h_t	Student hidden state at time t
m_t^i	Mastery logit for KC i at time t
μ_t^{agg}	Personalized knowledge aggregation
$G_t^{(h)}$	Dynamic Gaussian forgetting bias matrix
$\sigma_{t,h}$	Variance-derived decay parameter
O_t^{out}	Multi-scale fused student representation
$\hat{r}_{t+1}$	Performance prediction for next step

B Related Work

B.1 Knowledge Tracing

Early KT relied on Bayesian approaches (BKT) [13] which, despite their interpretability, were limited by the independence assumption among KCs. The paradigm shifted with DKT [42], employing RNNs [31] to model serialized knowledge states. Subsequent research has augmented this framework using Memory Networks [37], GNNs [26], and Transformers [48] to capture fine-grained fluctuations. Despite these advancements, most DLKT models rely on discrete ID embeddings, failing to encapsulate the latent semantic structures between KCs. Furthermore, to mitigate data sparsity, many approaches [8, 29, 32] oversimplify the task into a

one-to-one mapping between questions and KCs, obscuring the rich semantic information inherent in real-world exercises. To align knowledge structures, recent works have incorporated auxiliary side information. Graph-based methods, such as GKT [38], utilize knowledge graphs to characterize KC relationships, while methods like SKT [47] and SinKT [16] infer prerequisite relations via statistical methods or LLMs [3]. Although these approaches model dependencies, they typically treat them implicitly, neglecting the underlying semantic topology and hindering quantitative assessment of correlations. Additionally, handling questions with multiple KCs remains challenging. Existing solutions often employ naive fusion strategies (e.g., MLPs or average pooling) [8] to aggregate KC embeddings. This coarse-grained aggregation dilutes the unique identity of individual KCs and ignores structural dependencies.

Different from prior works that rely on rigid point embeddings, our proposed S^2KT framework addresses the heterogeneous distributional gap between Questions and KCs. We adopt a Gaussian Mixture Distribution paradigm to explicitly model the intrinsic uncertainty and continuous semantic topology of student learning.

B.2 Probabilistic embedding

Traditional representation learning relies on deterministic point vectors (e.g., Word2Vec [36]). While efficient, mapping entities to fixed Euclidean coordinates fails to capture inherent uncertainty, polysemy, and asymmetric hierarchies. To address this, research has shifted toward Probabilistic Embeddings, which represent entities as probability distributions. In the field of Natural Language Processing (NLP), probabilistic methods address polysemy and hierarchical entailment [49], utilizing GMMs to capture diverse semantics [4]. In Computer Vision (CV), they quantify uncertainty arising from noisy inputs [45] (e.g., occlusion) and facilitate many-to-many multimodal matching [12]. Similarly, in Knowledge Graphs, models like PERM [11] and Query2GMM [50] leverage the closed-form properties of Gaussians to execute complex first-order logical operations, such as handling logical “OR” queries. However, directly adapting these paradigms to KT presents two fundamental challenges:

- **Data Sparsity:** Existing methods rely on rich contexts or dense features to shape distributions. In contrast, KT exercises often consist of sparse text or mere IDs, lacking the semantic signal required to fit complex Gaussian models. This often leads to inaccurate parameter estimation or “distribution collapse,” negating the benefits of uncertainty modeling.
- **Temporal Dynamics:** While prior works focus on static entity ambiguity, KT aims to track the dynamic evolution of student “cognitive uncertainty.” This temporal non-stationarity is ill-suited for traditional static probabilistic models.

C Method Details

C.1 Implicit Relation Modeling

Since explicit dependency annotations are unavailable, we construct the relation edges via a two-step data-driven inference. First, we infer **KC-KC correlations** based on performance consistency. We hypothesize that semantic similarity manifests as consistent response patterns. The dependency weight S_{ij}^C between concepts

c_i and c_j is calculated as the probability of observing identical outcomes (simultaneous success or failure) within student histories:

$$S_{ij}^C = \frac{\sum_{u,t,t'} \mathbf{1}(r_u^t = r_u^{t'} \wedge c_u^t = c_i \wedge c_u^{t'} = c_j)}{\text{Count}(c_i, c_j)} \quad (27)$$

where $\mathbf{1}(\cdot)$ is the indicator function and noise is filtered via a threshold τ .

Subsequently, we propagate these micro-level dependencies to infer **Question-Question correlations**. Let $\mathbf{M} \in \{0, 1\}^{|Q| \times |C|}$ be the Question-KC association matrix. We derive the question similarity matrix S^Q by projecting questions into the learned concept manifold:

$$S^Q = \mathbf{D}^{-\frac{1}{2}} (\mathbf{M} \mathbf{S}^C \mathbf{M}^\top) \mathbf{D}^{-\frac{1}{2}} \quad (28)$$

where $\mathbf{D}$ is the degree matrix for normalization. This matrix operation effectively diffuses semantic information, establishing connections between questions with disjoint KC sets if their underlying concepts are structurally correlated.

D Datasets

Table 3: Statistics of the six datasets.

Dataset	Subject	# Students	# Interactions	# Questions	# KCs
Assistent09	Math	3,852	282,619	17,737	123
Assistent12	Math	46,674	6,123,270	179,999	265
AL2005	Math	574	609,979	173,650	112
Junyi	Math	191,874	25,852,548	721	39
Slepemapy	Geography	84,911	9,797,342	2,911	1,458
Ednet	English	735,190	95,190,437	12,283	189

ASSIST09. [15]³ Collected from the ASSISTments intelligent tutoring system, this dataset represents classic sparse K-12 mathematics interaction data. It is widely used for evaluating cold-start problems and multi-concept mapping tasks.

ASSIST12. [21]⁴ This dataset comprises K-12 mathematics exercise records spanning a full school year with timestamps, making it ideal for verifying long-term temporal dependencies.

Algebra05. [46]⁵ Derived from the KDD Cup 2010 challenge, this dataset records fine-grained, step-level interactions in algebra, which is effective for evaluating the validity of modeling cognitive processes.

Junyi. [7]⁶ A large-scale Chinese mathematics dataset from Junyi Academy. It includes expert-annotated prerequisite relation graphs, suitable for assessing models on knowledge structure and semantic relationships.

Slepemapy. [40]⁷ Originating from a geography practice system, this dataset focuses on factual knowledge retrieval. Its rich response time data allows for evaluating forgetting curves and memory decay mechanisms.

EdNet. [10]⁸ Containing over 95 million TOEIC learning records from the Santa app, this is one of the largest public educational datasets. It serves to test the scalability and robustness of models on industrial-grade heterogeneous data.

³<https://sites.google.com/site/assistmentsdata/home/assistent-2009-2010-data/combined-dataset-2009-10>

⁴<https://sites.google.com/site/assistmentsdata/datasets/2012-13-school-data-with-affect>

⁵<https://pslcdatashop.web.cmu.edu/KDDCup/downloads.jsp>

⁶<https://pslcdatashop.web.cmu.edu/DatasetInfo?datasetId=1198>

⁷<https://www.fi.muni.cz/adaptivelearning/data/slepemapy/>

⁸<https://github.com/rriid/ednet>

In preprocessing each dataset, we divide the response sequences of each student into subsequences, with each subsequence comprising up to 200 responses. Subsequences with fewer than 3 responses are discarded, while those shorter than 200 responses are padded with zeros to achieve the required length. The statistics for the processed datasets are presented in Table 3.

E Baselines & Implementation Details

E.1 Baselines

Sequential Methods. **DKT** [42] is a seminal approach utilizing RNNs to model evolving knowledge states as high-dimensional latent vectors. **DKT+** [52] adds reconstruction and waviness regularization terms to DKT to mitigate inconsistent predictions. **AT-DKT** [33] augments DKT with auxiliary tasks like question tagging and individualized prior knowledge prediction. **ATKT** [20] employs adversarial training with efficient perturbations to improve generalization on sparse data. **KQN** [28] models interactions via dot-product similarity between student knowledge and skill embeddings. **QIKT** [8] uses a dual-channel architecture to capture question-centric and concept-centric representations via an IRT-based layer.

Graph-Based Methods. **GKT** [38] models knowledge structure as a graph, propagating influence across neighboring concepts using GNNs. **SKT** [47] models knowledge tracing as influence propagation to capture both temporal effects and spatial relations. **PEBG** [32] constructs a question-skill bipartite graph to learn distinct question embeddings incorporating side information. **SinKT** [16] leverages LLMs to construct heterogeneous concept-question graphs for inductive predictions on unseen questions.

Memory-Augmented Methods. **DKVMN** [54] utilizes a static key matrix for concepts and a dynamic value matrix for evolving mastery. **SKVMN** [1] integrates a Hop-LSTM into DKVMN to capture sequential dependencies between memory accesses. **Deep-IRT** [51] synthesizes DKVMN with Item Response Theory to explicitly estimate student ability and item difficulty parameters.

Attention-Based Methods. **SAKT** [39] is the first self-attentive model adapting Transformers to weigh relevant past interactions. **AKT** [17] employs monotonic attention and context-sensitive embeddings to model the decay of information relevance. **SimpleKT** [34] combines standard dot-product attention with Rasch-model-based

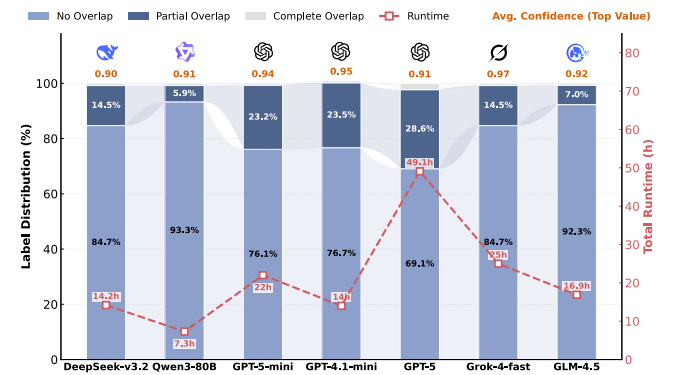


Figure 6: Performance comparison of LLMs on KC semantic inference.

question embeddings. **DTransformer** [53] introduces a contrastive learning objective to ensure stability in diagnostic tracing. **FlucKT** [22] uses decomposition-based causal convolution to disentangle long-term trends from short-term fluctuations. **FoLiBiKT** [24] incorporates a linear bias term to explicitly penalize attention to distant interactions, modeling forgetting. **SparseKT** [23] implements k -sparse attention to select only the top- k most relevant interactions, filtering noise. **StableKT** [30] employs linear biases and multi-head aggregation to maintain stability across long interaction sequences. **RobustKT** [19] decouples true cognitive patterns from error-prone data to enhance reliability in noisy environments.

E.2 Implementation Details

To ensure a fair comparison, baseline hyperparameters are initialized according to their original papers and carefully fine-tuned on our benchmark. For S^2KT , we optimize hyperparameters via grid search: the learning rate is selected from $\{1e^{-4}, 5e^{-4}, 1e^{-3}, 5e^{-3}, 1e^{-2}\}$, and the latent dimension d is chosen from $\{16, 32, 64, 128\}$. We use the Adam optimizer with a batch size of $\{64, 128, 256\}$. The training process runs for a maximum of 200 epochs, with an *early stopping* strategy of 10 epochs based on the validation AUC to prevent overfitting.

F Supplements For Experiments

F.1 Empirical Analysis of LLM-driven Semantic Inference

To assess how well LLMs capture semantic relations between KCs in a KT context. We first filtered the original KC set to retain only those with textual annotations (to leverage the LLMs’ language understanding). From this filtered set, we formed all pairs of KCs and defined three simplified semantic relation categories between any two KCs: **Full Coverage** (one KC’s content fully entails the other), **Partial Coverage** (they share some content overlap), and **No Coverage** (no semantic overlap). For each KC pair, we prompted several different LLMs to judge the relation category using the standardized instruction template detailed in Table 4. We also recorded the inference time each model required to evaluate the entire dataset. The results, as illustrated in Figure 6, highlight three key observations: **(1) High Confidence via Prompt Engineering:** The LLMs consistently reported very high confidence scores on this task (mean ≈ 9.3 out of 10). This indicates that our prompts effectively guided the models to make self-consistent inferences and to quantify their certainty. **(2) Model Variance and Data Bias:** We observed substantial differences in the distribution of predicted relation categories across different LLMs. For instance, one model family might label many KC pairs as “partial coverage,” whereas another might seldom do so. In contrast, models within the same family or training paradigm produced very similar relation distributions. This pattern strongly suggests that the discrepancies arise from the models’ training data, not their architectures. Indeed, it is well-known that LLMs inherit biases and skewed patterns from their pretraining corpora. In KT, such “intrinsic” data bias may have led one LLM to perceive more semantic overlap between KCs than another, which in turn could bias any downstream KT model that relies on these semantic judgments. **(3) High Computational**

Cost: Although the dataset was small (123 KCs), LLM-based inference proved extremely time-consuming, taking on the order of 7–49 hours per model. For example, even the relatively efficient Qwen3-80B required 7.3 hours, **whereas the computationally intensive GPT-5 took nearly 49.1 hours** to process the same pairs. This highlights a significant practical bottleneck: utilizing LLMs for fine-grained semantic analysis in KT is resource-intensive.

Table 4: The detailed prompt template for semantic relationship labeling. The prompt utilizes coarse-grained semantic categories and confidence scores to resolve the representational heterogeneity among KC texts.

Component	Specification Detail
System Role	Function as an educational domain expert. Determine relationships based on <i>intrinsic semantic meaning</i> rather than surface lexical matching.
Criteria	1. Complete Overlap: Concepts are semantically interchangeable or minor wording variations. 2. Partial Overlap: Concepts share a logical intersection (e.g., <i>Definition-Example</i>, or shared sub-scope) but contain distinct info. 3. No Overlap: Concepts address disjoint topics.
Guidelines	• Treat mathematical formulas as core semantic content. • Handle ambiguity by selecting the plausible label while lowering the confidence score. • Strictly adhere to the JSON format constraints.
Few-Shot Example	Text for ID_1: “Commutative Property” Text for ID_2: “Definition of Slope” Output: <code>{"id1": "ID1", "id2": "ID2", "label": "No Overlap", "explanation": "Text 1 describes an algebraic property, while Text 2 defines a geometric concept.", "confidence": 0.97}</code>
Input	Text for ID_1: {Concept_Text_1} Text for ID_2: {Concept_Text_2}
Output	JSON Object: {id1, id2, label, explanation, confidence}